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Overview: This document details the step-by-step assembly process of a pressure chamber as observed in the video. The process involves mounting internal components, aligning structural parts, and securing them with screws using hand tools.

Tools and Components Used:

- Allen wrenches
- Cylindrical base plate (metal)
- Middle spacer rings (with through-holes)
- Top flange/lid
- Set of evenly spaced fastener screws

Assembly Steps:

Step 1: Initial Layout and Preparation

- All components are laid out cleanly on the assembly platform.
- The base plate is placed at the center.
- A visual inspection is conducted to confirm presence of necessary parts.

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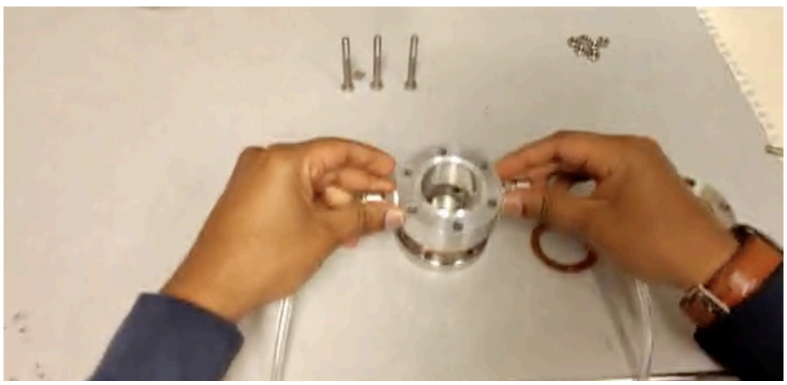
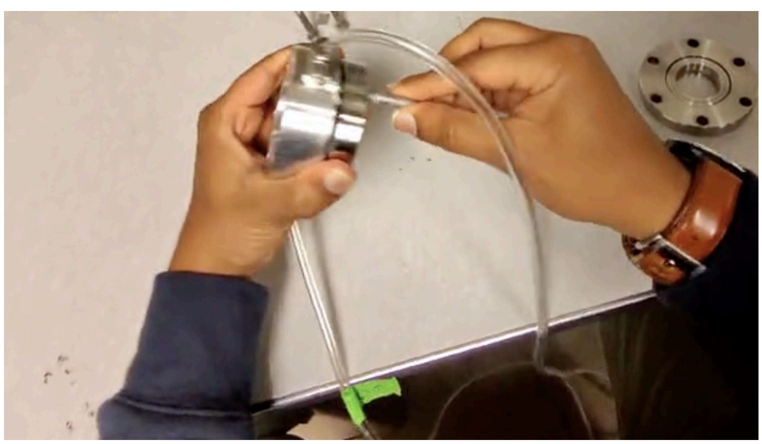
Step 2: Placement of First Spacer Cylinder

- The first cylindrical spacer is carefully aligned with the holes on the base plate.
- Orientation is checked manually to ensure that all mounting holes are matched.



Step 3: Fastening the First Spacer

- Screws are inserted through the spacer into the baseplate.



Step 4: Placement of O-Ring

- A mid-section component O-ring is placed on the spacer cylinder

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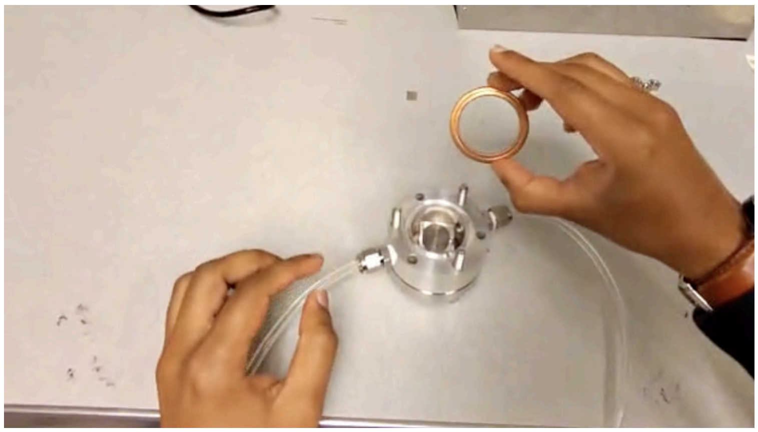
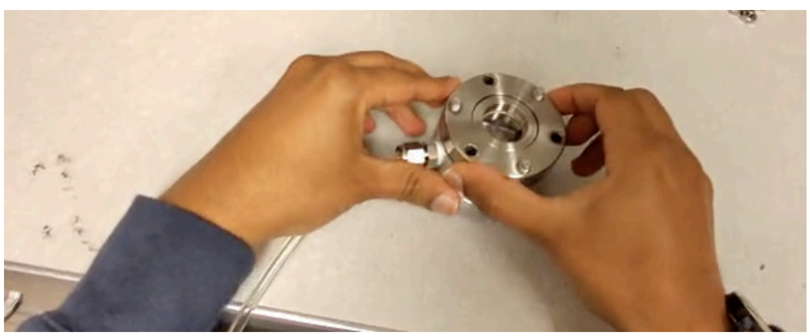


Step 5: Top Flange Positioning

- The topmost component (possibly a lid or top flange) is positioned.
- This part has evenly spaced screw holes around the perimeter.
- It is gently lowered onto the previously assembled stack.



Step 6: Sequential Fastening of Top Flange

- Screws are placed through the top flange into the underlying threaded holes.
- A precise, diagonal tightening sequence is followed to prevent tilt or misalignment.

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Step 7: Final Tightening and Structural Inspection

- All fasteners are revisited in a loop to ensure uniform tightness.

Step 9: Completion Check

- Visual overview confirms that the chamber is fully assembled.
- All screws are tightened, and all surfaces appear clean.
- No loose parts or visible defects are observed.

Recommendations for Replication:

- Ensure all screws are of uniform length and thread type.
- Clean all contact surfaces before assembly to avoid contamination.
- Use calibrated torque tools if assembling for vacuum or high-pressure environments.
- Document part serial numbers for traceability if necessary.

End of Documentation